Power Tree App

User Manual

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1. Introduction

Power Tree App is a browser-based tool for modeling the power consumption of electronic devices. It provides a drag-and-drop canvas where you build a power tree by placing and connecting **power sources**, **converters** (switching or LDO), **series elements** (load switches, FETs, diodes), and **loads**. The app automatically calculates power flow, losses, and efficiency at every node in real time.

Key capabilities include:

- Real-time power analysis with voltage, current, and loss displayed directly on the canvas.
- Multiple **power states** (e.g., Active, Sleep) with independent load configurations and weighted-average analysis.
- Min/Nominal/Max voltage scenario analysis for multi-rail designs.
- Battery lifetime estimation with temperature-dependent capacity and discharge curves.
- Screenshot-based graph digitizer for extracting efficiency and discharge curves from datasheets.
- Full project save/load in JSON format with undo/redo history.

2. Getting Started

2.1 Launching the App

Open the app in a modern web browser (Chrome, Edge, Firefox, or Safari recommended). The app runs locally via Vite at `http://localhost:5173/` by default. No installation or login is required.

2.2 UI Overview

The interface is divided into the following regions:

- **Sidebar (left):** Lists the six draggable component types. Drag any item onto the canvas to place it. A theme toggle (Light/Dark) is at the bottom of the sidebar.
- **Toolbar (top):** Contains the editable *project name*, and buttons for *Load*, *Save*, *Save As*, and *Details* (analysis results).
- **State Tabs (below toolbar):** Tabs for each power state, plus a *Manage States* button.
- **Canvas (center):** The interactive workspace where you build your power tree.
- **Config Panel (right):** Appears when a node is selected; shows all configuration options for that node.

3. Canvas Interactions

3.1 Pan and Zoom

- **Pan:** Left-click and drag on empty canvas space to pan.
- **Zoom:** Scroll wheel to zoom in/out. Zoom range is 5% to 400%.
- **Fit view:** Use the built-in controls (bottom-left corner) to fit all nodes in view.

3.2 Selecting and Moving Nodes

- **Single select:** Click a node to select it. Its configuration panel appears on the right.
- **Multi-select:** Hold **Shift** and left-click drag to draw a selection box (lasso). All nodes partially inside the box are selected.
- **Move:** Drag any selected node to reposition it. If multiple nodes are selected, they move together.
- **Deselect:** Click empty canvas space to deselect all nodes and close the config panel.

3.3 Drawing Connections

Each power node has circular **handles** on its edges. Source nodes have a handle on the right; converters and series elements have handles on both left (input) and right (output); loads have a handle on the left (input). The connection radius is generous (30px) so you don't need to be pixel-precise.

- Drag from a source handle (right side) to a target handle (left side) to create a connection.
- Connections are rendered as smooth bezier curves with automatic collision avoidance — edges route around intervening nodes when possible.
- Each node can have one parent (input) and multiple children (outputs).

3.4 Deleting Elements

- Press **Backspace** to delete all selected nodes and edges.
- Individual nodes can also be deleted via the **Delete** button in the config panel.
- Deleting a node also removes all edges connected to it.

4. Components

Drag components from the sidebar onto the canvas. Click any placed node to open its configuration panel.

4.1 Power Source

Represents the input power supply: a battery, USB port, wall adapter, or any DC source.

Field	Description
Label	Display name for the source (e.g., "3.7V Li-Po").
Source Type	**Fixed Supply** or **Battery**. Selecting Battery unlocks battery-specific fields (see Section 9).
Nominal Voltage (V)	The primary operating voltage.
Min Voltage (V)	Optional. If set, enables a Min Vin analysis scenario.
Max Voltage (V)	Optional. If set, enables a Max Vin analysis scenario.
Internal Resistance (mΩ)	Source internal resistance. Causes I^2R power dissipation within the source.

4.2 Converter

Represents a voltage regulator: switching (buck, boost) or LDO.

Field	Description
Label	Display name (e.g., "TPS62840").
Enabled	ON/OFF toggle. When OFF, the converter and everything downstream is disconnected (no current flows).
Type	**Switching** or **LDO**.
Output Voltage (V)	The regulated output voltage.
Quiescent Current (µA)	Baseline current draw of the converter itself (added to the total input current).

Efficiency Curves (Switching Converters Only)

Switching converters use efficiency curves — **efficiency (%) vs. load current (A)** — to determine losses at each operating point. You can provide separate curves for different input voltages.

For each Vin curve tab, you can enter efficiency data three ways:

1. **Manual entry:** Add rows of Load (A) / Efficiency (%) pairs.
2. **CSV upload:** Import a CSV file with columns for load current and efficiency.
3. **Screenshot extraction:** Upload a datasheet screenshot and digitize the curve (see Section 6).

LDO converters do not use efficiency curves. Their efficiency is computed automatically as `Vout / Vin`, where Vin is inherited from the parent node.

4.3 Series Element

Represents an inline component such as a load switch, MOSFET, fuse, or diode.

Field	Description
Label	Display name (e.g., "TPS22918").
Enabled	ON/OFF toggle. When OFF, acts as an open switch — everything downstream is disconnected.
Mode	**Resistor/FET** (Rdson) or **Diode** (Vf).
Rdson (mΩ)	On-resistance (Resistor/FET mode). Power loss = I^2R. Voltage drop = I × R.
Vf (mV)	Forward voltage drop (Diode mode). Power loss = I × Vf.

4.4 Load

Represents a power-consuming component: an MCU, sensor, LED, resistor, etc. The input voltage is automatically inherited from the parent node.

Field	Description
Label	Display name (e.g., "nRF52840").
Enabled	ON/OFF toggle. When OFF, the load draws zero current.
Load Mode	One of three modes (see below).

Load Modes

- **Fixed Current:** A single constant current value (entered in mA). Useful for simple, steady loads.
- **Current Profile (vs. Time):** A periodic time-series of current values. Define time (s) and current (A) pairs in a table, or upload a CSV file. The profile repeats periodically; the period equals the last time entry.
- **Resistor:** A constant resistance value (in Ω). Power is calculated as V^2/R. The resistance is *global* and does not change between power states.

> **Note:** The load's label, load mode, and resistance value persist across power states. Only the current

> profile / fixed current value changes per state.

4.5 Box (Group Outline)

A resizable rectangular outline for visually grouping nodes on the canvas. Boxes are purely cosmetic and do not affect power analysis.

- Drag "Box" from the sidebar to place.
- Select the box to reveal resize handles on corners and edges; drag to resize.
- Boxes have the *lowest* click priority: clicking a node or edge that overlaps a box will select the node/edge, not the box.
- Box dimensions are saved with the project file.

4.6 Text Label

A free-floating, draggable text annotation. Double-click to edit; press Enter to confirm (Shift+Enter for a new line). When selected, a small config area appears with:

- **Font size** (8–72 pt).
- **Color** presets via swatches.
- **Delete** button.

5. Auxiliary Loads

Sources, converters, and series elements support **auxiliary loads** — small, persistent current draws that don't warrant a full load node. Common examples include pull-up resistors, resistor dividers, indicator LEDs, and feedback networks.

Adding Auxiliary Loads

In the config panel for any source, converter, or series element, find the **Auxiliary Loads** section. Click the **+ Add** button (always visible in the section header) to add an entry.

Field	Description
Name	A label for the auxiliary load (e.g., "Pull-up R1").
Mode	**R** (resistor, in Ω) or **I** (fixed current, in mA).
Value	Resistance in ohms or current in milliamps, depending on mode.

Auxiliary loads are **global by default** — they draw current in every power state. However, each auxiliary load has a per-state **ON/OFF toggle**. Switching it off in a particular state means that load does not contribute current in that state's analysis.

6. Graph Digitizer (Screenshot Extraction)

The graph digitizer lets you extract numeric data from a screenshot of a graph — typically an efficiency curve from a converter datasheet or a discharge curve from a battery datasheet.

How to Use

1. **Upload an image** of the graph (PNG, JPG, etc.).
2. **Set Point 1 (P1):** Click the bottom-left corner of the plot area. Enter the corresponding X and Y axis values.
3. **Set Point 2 (P2):** Click the top-right corner of the plot area. Enter the corresponding X and Y axis values.
4. **Trace the curve:** Click along the curve to add data points. The digitizer maps pixel positions to data coordinates using the P1/P2 calibration.
5. **Apply Points:** Click to confirm (requires at least 2 traced points). Points are sorted by X and merged if within 5% relative distance.

> **Tip:** Enable **Log X** for efficiency curves where the x-axis (load current) uses a logarithmic scale. This is the default when extracting converter efficiency curves.

Use **Undo** to step back (removing the last traced point, then corners, then P1). Use **Reset** to start over entirely.

7. Power States

Power states model different operating modes of your device (e.g., Active, Sleep, Transmit, Idle). Each state has its own load configurations, and the overall analysis computes a time-weighted average across all states.

7.1 Switching States

Click a state tab below the toolbar to switch to that state. The canvas updates to show each load's configuration (current profile, enabled status) for the selected state. When you modify a load's current or toggle a node on/off, the change applies *only* to the currently active state.

7.2 Managing States

Click **Manage States** to open the state manager overlay, where you can:

- **Rename** a state.
- Set the **time fraction (%)** — the percentage of total operating time spent in this state. A warning appears if fractions don't sum to 100%.

- **Copy** a state to duplicate it (including all load snapshots and enabled overrides) as a starting point for a new state.
- **Remove** a state (at least one state must remain).
- **Add** a new state by entering a name.

7.3 Per-State Enable/Disable

The **enabled/disabled** toggle on converters, series elements, and loads is saved independently per power state. For example, you can disable a sensor's converter in Sleep mode while keeping it enabled in Active mode. When you switch states, the enabled status of every node is restored to match that state's configuration.

8. Analysis & Results

8.1 Automatic Analysis

The power analysis runs automatically whenever you change the power tree (add/remove/configure nodes, draw/delete connections, switch states). There is no "Analyze" button — results update in real time.

8.2 Node Badges

Each node displays an analysis badge showing key metrics for the active power state:

Node Type	Badge Contents
Source	Power In, Power Out. Battery lifetime if applicable.
Converter	Power In, Power Out, Efficiency (%).
Series Element	Power In, Power Out, Loss (mW).
Load	Power In.

Series elements also display the **voltage drop** (Vdrop) on the node body. Disabled nodes show grayed-out styling with "0 mA" labels.

8.3 Edge Current Labels

Each connection line displays the **average current** flowing through it (in mA or A). This is the time-weighted average across the load's periodic profile. Edges connected to disabled nodes show "0 mA" and are rendered with a dashed, dimmed style without animation.

8.4 Results Panel

Click **Details** in the toolbar to open the full results panel. It contains:

- **Summary cards:** Average input power, average load power, total loss, and system efficiency.
- **Battery lifetime** cards (one per battery source, if applicable).
- **Power loss by component:** A donut chart with a legend showing each component's power loss in mW.
- **Power over time:** Line chart of input power and load power vs. time (if dynamic loads exist).
- **Input current over time:** Line chart of total input current vs. time.
- **Battery discharge:** Voltage vs. time chart for each battery source with detailed discharge curves.
- **Per-node table:** Tabular breakdown of input power, output power, loss, and efficiency for every component.

8.5 Voltage Scenarios (Min/Nom/Max)

If any power source has a **Min** or **Max** voltage defined, the results panel shows scenario tabs: **MIN Vin**, **NOM Vin**, and **MAX Vin**. Each tab displays the full analysis recalculated with that source voltage. This lets you evaluate worst-case and best-case power budgets.

9. Battery Modeling

When a power source is set to **Battery** mode, additional configuration fields appear for modeling battery capacity and estimating operating lifetime.

9.1 Simple Mode

In Simple mode, you define a table of **temperature (°C)** and **capacity (mAh)** pairs. The engine interpolates capacity based on your operating temperature profile to compute an effective capacity.

This mode is suitable when you have a single nominal capacity rating at different temperatures from the battery datasheet and don't need to model the full discharge curve.

9.2 Detailed Mode

In Detailed mode, you provide full **voltage vs. capacity** discharge curves, one per temperature. Each temperature tab lets you enter data points manually, upload a CSV, or use the graph digitizer to extract the curve from a datasheet screenshot.

The engine simulates discharge step-by-step: as the battery depletes, the voltage changes according to the discharge curve, which in turn affects converter efficiencies and current draw. This produces an accurate, non-linear lifetime estimate and generates a **voltage vs. time** discharge chart in the results panel.

9.3 Battery Lifetime Calculation

Both modes require an **operating temperature profile** — a table of temperature (°C) and time fraction (%) pairs that should sum to 100%. The engine uses this to weight capacity across temperatures.

- **Simple mode:** Lifetime (hours) = Effective Capacity (mAh) / Average Current (mA).
- **Detailed mode:** Lifetime is computed via step-by-step discharge simulation down to the **cutoff voltage**.

Battery lifetime accounts for all power states: the average current is the time-weighted average across all states.

A **cutoff voltage** field specifies the minimum operating voltage; the battery is considered depleted when voltage drops to this level.

10. Project Management

10.1 Save & Load

Saving

- **Save:** Saves the project to the previously used file location (if one exists). If this is a new project that has never been saved, behaves like Save As. A brief "Saved" toast confirmation appears on success.
- **Save As:** Always prompts for a file location (or downloads a `.json` file if the File System Access API is not available).
- **Cmd+S / Ctrl+S:** Keyboard shortcut for Save.

Loading

- Click **Load** to open a file picker. Select a previously saved `.json` project file.
- The app remembers the loaded file's location, so subsequent **Save** operations write back to the same file without prompting.

The project file stores all nodes (positions, data, dimensions), edges, power states (including load snapshots, enabled overrides, and auxiliary load overrides), the project name, and the theme preference.

10.2 Undo & Redo

- **Cmd+Z / Ctrl+Z:** Undo the last change.
- **Cmd+Shift+Z / Ctrl+Y:** Redo.
- History stores up to **50** snapshots of the complete graph and power state configuration.

10.3 Copy, Cut & Paste

- **Cmd+C / Ctrl+C:** Copy selected nodes to clipboard. Edges are included only if *both* endpoints are selected.
- **Cmd+X / Ctrl+X:** Cut — copies selected nodes then deletes them from the canvas.
- **Cmd+V / Ctrl+V:** Paste — places copied nodes at a +40px offset with new IDs. Internal edges between pasted nodes are preserved. *No automatic connections are drawn to existing nodes.*

> **Note:** Keyboard shortcuts (except Undo/Redo) are disabled when focus is in a text input field to avoid interfering with normal typing.

10.4 Theme (Light / Dark)

Toggle between light and dark themes using the button at the bottom of the sidebar. The theme preference is saved with the project file and persists across sessions.

11. Keyboard Shortcuts

Shortcut	Action
Cmd/Ctrl + Z	Undo
Cmd/Ctrl + Shift + Z *or* Cmd/Ctrl + Y	Redo
Cmd/Ctrl + C	Copy selected nodes
Cmd/Ctrl + X	Cut selected nodes
Cmd/Ctrl + V	Paste
Cmd/Ctrl + S	Save project
Backspace	Delete selected nodes/edges
Shift + Left-click drag	Lasso (box) select
Scroll wheel	Zoom in/out

12. Calculation Engine Details

This section describes how the engine computes power flow for advanced users who want to understand the underlying math.

12.1 Converter Efficiency

Switching Converters

The engine looks up efficiency from the provided curves. If multiple Vin curves exist, it interpolates between the two nearest Vin curves. Within a single curve, it interpolates linearly between the two nearest load current points. If the operating point is outside the curve range, the nearest endpoint value is used.

Input current is calculated as: `I_in = (I_out × V_out) / (V_in × η) + I_q`, where η is the interpolated efficiency and I_q is the quiescent current.

LDO Converters

LDO efficiency is defined as `η = V_out / V_in`. Input current equals output current plus quiescent current: `I_in = I_out + I_q`.

12.2 Series Element Loss

- **Resistor/FET mode:** Power loss = $I^2 \times R_{dson}$. The voltage downstream of the element is reduced by $I \times R_{dson}$.
- **Diode mode:** Power loss = $I \times V_f$. The voltage downstream is reduced by V_f.

Internal resistance on a power source works similarly: power dissipated in the source = $I^2 \times R_i$.

12.3 Unified Timeline & Averaging

When multiple loads have periodic current profiles with different periods, the engine must compute a time-weighted average over a common interval. It uses a three-tier approach:

1. **Exact LCM:** If the Least Common Multiple of all load periods produces ≤ 2,000 time points, it is used for perfect accuracy.
2. **Pseudo-LCM with snapping:** If the exact LCM is too large, the engine attempts to quantize each period to the nearest multiple of a common resolution, such that all snapped periods are within **5%** of their original values. If this yields ≤ 2,000 points, it is used.
3. **Uniform sampling:** As a fallback, 1,000 uniformly spaced points are sampled over the longest individual period. The engine's modular arithmetic ensures correct periodic interpolation.

All power and current values are then computed at each time point and averaged using trapezoidal time-weighting across the unified timeline.

13. Tips & Best Practices

- **Start from the source:** Place your power source first, then add converters and loads from left to right to match the natural power flow direction.
- **Use auxiliary loads for small draws:** Pull-up resistors, feedback dividers, and indicator LEDs don't need their own load node — add them as auxiliary loads on the relevant converter or source.
- **Copy power states:** When creating similar states (e.g., Active TX vs. Active RX), copy an existing state and modify only the loads that differ.
- **Use boxes and text labels:** Group related subsystems with Box outlines and add Text labels for documentation directly on the canvas.
- **Check scenario tabs:** If your source voltage varies (e.g., a battery from 4.2V to 3.0V), define min/max voltages to see worst-case efficiency and power.
- **Save frequently:** Use Cmd+S / Ctrl+S to save. The app stores the complete project state including all power states and analysis configuration.
- **Use the graph digitizer for datasheet curves:** Instead of manually entering dozens of efficiency data points, screenshot the datasheet graph and trace it with a few clicks.
- **Keep load time series simple:** Fewer time steps in a load profile means faster analysis. Use Fixed Current mode when a load has a constant draw.